

Summer Term 2026

Investment and Financial Management

Introduction to Corporate Finance

Sessions 09+10: Capital structure

Berk/DeMarzo (2024): Corporate Finance, 6th edition, Pearson.

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- 1. Equity vs. debt financing**
 - 2. Modigliani-Miller I: Leverage and arbitrage**
 - 3. Modigliani-Miller II: Leverage and cost of capital**
 - 4. Capital structure fallacies**
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1. Equity vs. debt financing

2. Modigliani-Miller I: Leverage and arbitrage

3. Modigliani-Miller II: Leverage and cost of capital

4. Capital structure fallacies

- **Basic question of corporate finance = Capital structure decision**

After the decision whether or not to undertake an investment, which type of security should a firm sell to investors resp. which capital structure should be decided on?

- **Definition of capital structure**

The relative proportions of debt, equity, and other securities that a firm has outstanding. The two main types of capital structure are:

- **Financing a firm with equity** (= all-equity financing, unlevered firm)
- **Financing a firm with equity and debt** (= levered firm)



Is there a capital structure that shareholders prefer?

1. Equity vs. debt financing

2. Modigliani-Miller I: Leverage and arbitrage

3. Modigliani-Miller II: Leverage and cost of capital

4. Capital structure fallacies

- **Capital structure decision:** The theory of capital structure was developed beginning with the capital structure theory of Merton Miller and Franco Modigliani (MM).

- **Assumptions of the MM model:**
 - **Homogeneous expectations**
 - **Homogeneous business risk classes**
 - **Perpetual cash flows**
 - **Perfect capital markets:**
 - Investors and firms can trade the same set of securities at competitive market prices equal to the present value of their future cash flows.
 - There are no taxes, transaction costs, issuance costs or agency costs associated with security trading.
 - A firm's financing decisions do not change the cash flows generated by its investments, nor do they reveal new information about them.

- **Modigliani and Miller argue:**

In a perfect capital market, the total value of a firm is equal to the market value of the total cash flows generated by its assets and is not affected by its choice of capital structure. Reasons:

- The market value of any firm is independent of its capital structure.
- If a company has a given set of assets, changing debt to equity will change the way net operating income is divided between lenders and shareholders, but will not change the value of the company.

MM proposition I (without taxes):

The market value of a company is not affected by the capital structure of the company.

$$E + D = U = A$$

E = Market value of equity in a levered firm

D = Market value of debt in a levered firm

U = Market value of equity in an unlevered firm

A = Market value of the firm's assets

▪ Proof of MM proposition I:

- **Law of one price:** The firm's securities and its assets need to have the same total market value.
 - Otherwise, an arbitrage opportunity would exist.
- **Homemade leverage:** The investors use leverage in their own portfolios to adjust the leverage choice made by the firm.
 - Homemade leverage is a perfect substitute for corporate leverage
 - Substitutability between corporate debt and personal debt

▪ Law of one price: Equity and debt financing

Problem

- You are considering an investment opportunity. For an initial investment of \$800 this year, the project will generate cash flows of either \$1400 or \$900 next year, depending on whether the economy is strong or weak, respectively. Both scenarios are equally likely. The risk-free rate is 5%, the risk premium is 10% and the cost of capital for this project is 15%. Suppose, the entrepreneur borrows \$200 when financing the project.
- According to Modigliani and Miller, what should be the value of the equity? What is the expected return?

- **Law of one price: Equity and debt financing**

Solution

Solution

Because the value of the firm's total cash flows is still \$1000, if the firm borrows \$200, its equity will be worth \$800. The firm will owe $\$200 \times 1.05 = \210 in one year. Thus, if the economy is strong, equity holders will receive $\$1400 - \$210 = \$1190$, for a return of $\$1190/\$800 - 1 = 48.75\%$. If the economy is weak, equity holders will receive $\$900 - \$210 = \$690$, for a return of $\$690/\$800 - 1 = -13.75\%$. The equity has an expected return of

$$\frac{1}{2}(48.75\%) + \frac{1}{2}(-13.75\%) = 17.5\%.$$

Note that the equity has a return sensitivity of $48.75\% - (-13.75\%) = 62.5\%$, which is $62.5\%/50\% = 125\%$ of the sensitivity of unlevered equity. Its risk premium is $17.5\% - 5\% = 12.5\%$, which is also 125% of the risk premium of the unlevered equity, so it is appropriate compensation for the risk.

- **Homemade leverage and arbitrage**

Homemade Leverage and Arbitrage

Problem

Suppose there are two firms, each with date 1 cash flows of \$1400 or \$900 (as shown in Table 14.1). The firms are identical except for their capital structure. One firm is unlevered, and its equity has a market value of \$990. The other firm has borrowed \$500, and its equity has a market value of \$510. Does MM Proposition I hold? What arbitrage opportunity is available using homemade leverage?

■ Homemade leverage and arbitrage

Solution

MM Proposition I states that the total value of each firm should equal the value of its assets. Because these firms hold identical assets, their total values should be the same. However, the problem assumes the unlevered firm has a total market value of \$990, whereas the levered firm has a total market value of \$510 (equity) + \$500 (debt) = \$1010. Therefore, these prices violate MM Proposition I.

Because these two identical firms are trading for different total prices, the Law of One Price is violated and an arbitrage opportunity exists. To exploit it, we can borrow \$500 and buy the equity of the unlevered firm for \$990, re-creating the equity of the levered firm by using homemade leverage for a cost of only $\$990 - \$500 = \$490$. We can then sell the equity of the levered firm for \$510 and enjoy an arbitrage profit of \$20.

	Date 0	Date 1: Cash Flows	
	Cash Flow	Strong Economy	Weak Economy
Borrow	\$500	−\$525	−\$525
Buy unlevered equity	−\$990	\$1400	\$900
Sell levered equity	\$510	−\$875	−\$375
Total cash flow	\$20	\$0	\$0

Note that the actions of arbitrageurs buying the unlevered firm and selling the levered firm will cause the price of the unlevered firm's stock to rise and the price of the levered firm's stock to fall until the firms' values are equal and MM Proposition I holds.

- **A leveraged recapitalization: Application of MM proposition I**
 - How do firms adjust their capital structure?
 - For instance, a firm can use borrowed funds to pay a large special dividend or repurchase a significant amount of outstanding shares.

▪ Leveraged recapitalization

Problem

- Harrison Industries is currently an all-equity firm operating in a perfect capital market, with 50 million shares outstanding that are traded for \$4 per share.
- Harrison plans to increase its leverage by borrowing \$80 million and using the funds to repurchase 20 million of its outstanding shares.
- This transaction can be viewed in two stages:
 - First, Harrison sells debt to raise \$80 million in cash.
 - Second, Harrison uses the cash to repurchase shares.

▪ Leveraged recapitalization

Solution (I)

Initial		After Borrowing	
Assets	Liabilities	Assets	Liabilities
		Cash	Debt
		80	80
Existing assets	Equity	Existing assets	Equity
200	200	200	200
200	200	280	280
Shares outstanding (million)	50	Shares outstanding (million)	50
Value per share	\$4.00	Value per share	\$4.00

- **Initial:**
 - Market value of Harrison's equity = 50 million shares x \$4 per share = \$200 million
- **After borrowing:**
 - Harrison's liabilities grow by \$80 million.

▪ Leveraged recapitalization

Solution (II)

Initial		After Borrowing		After Share Repurchase	
Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
		Cash	Debt	Cash	Debt
		80	80	0	80
Existing assets	Equity	Existing assets	Equity	Existing assets	Equity
200	200	200	200	200	120
200	200	280	280	200	200
Shares outstanding (million)	50	Shares outstanding (million)	50	Shares outstanding (million)	30
Value per share	\$4.00	Value per share	\$4.00	Value per share	\$4.00

• Share repurchase:

- Harrison spends the \$80 million in borrowed cash to repurchase 20 million shares ($\$80 \text{ million} \div \$4 \text{ per share} = 20 \text{ million shares}$)
- Decrease of assets by \$80 million, debt remains unchanged, market value of the equity must also fall by \$80 million, from \$200 million to \$120 million, for assets and liabilities to remain balanced.
- Share price unchanged: $\$120 \text{ million} \div 30 \text{ million shares} = \4 per share

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▪ Leverage and the equity cost of capital

- **Basis of MM II:** MM's first proposition can be used to derive an explicit relationship between leverage and the equity cost of capital. MM proposition I states that:

$$E + D = U = A$$

- Recall: The total market value of the firm's securities is equal to the market value of its assets, whether the firm is unlevered or levered.
- Thus, the return on unlevered equity (r_U) is related to the returns of levered equity (r_E) and debt (r_D):

$$r_U = \frac{E}{D + E} r_E + \frac{D}{D + E} r_D$$

MM proposition II (without taxes):

The cost of capital of levered equity is equal to the cost of capital of unlevered equity plus a premium that is proportional to the market value of the debt-to-equity ratio.

$$r_E = r_U + \frac{D}{E} (r_U - r_D)$$

r_E = Return on levered equity

r_U = Return on unlevered equity

r_D = Return on debt

D/E = Debt-to-equity ratio (market value)

- **Cost of capital of levered equity**

Computing the Equity Cost of Capital

Problem

Suppose the entrepreneur of Section 14.1 borrows only \$200 when financing the project. According to MM Proposition II, what will be the firm's equity cost of capital?

- **Cost of capital of levered equity**

Solution

Because the firm's assets have a market value of \$1000, by MM Proposition I the equity will have a market value of \$800. Then, using Eq. 14.5,

$$r_E = 15\% + \frac{200}{800}(15\% - 5\%) = 17.5\%$$

This result matches the expected return calculated in Example 14.1.

- **Unlevered firm: Capital budgeting and the weighted average cost of capital (WACC)**
 - If a firm is unlevered, all of the free cash flows generated by its assets are paid out to its equity holders.
 - The market value, risk, and cost of capital for the firm's assets and its equity coincide. Therefore:

$$r_U = r_A$$

- **Levered firm: Capital budgeting and the weighted average cost of capital (WACC)**

- If a firm is levered, project r_A is equal to the firm's weighted average cost of capital (r_{WACC}):

$$r_{WACC} = r_U = r_A$$

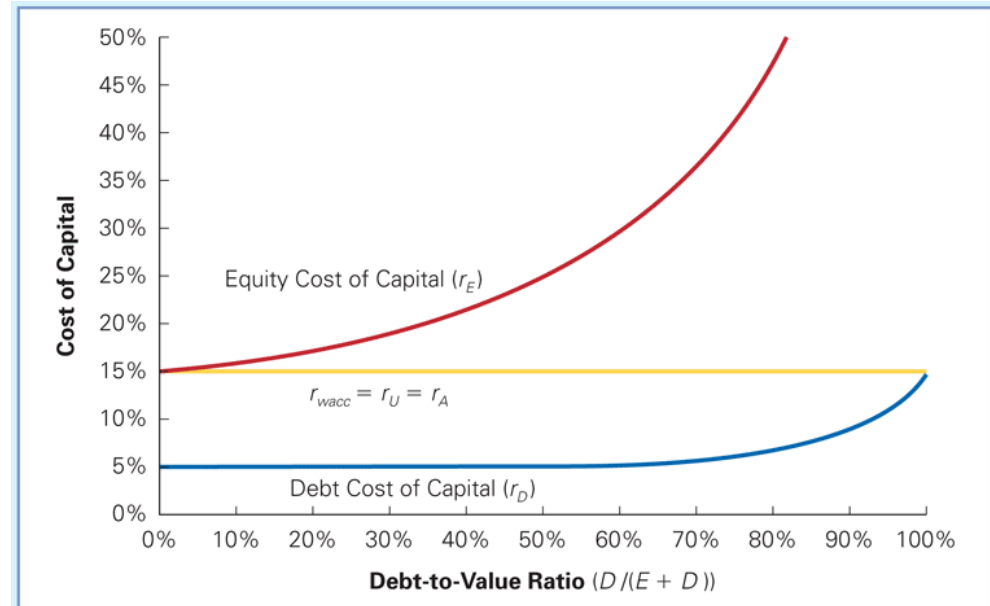
$$\begin{aligned} r_{WACC} = r_U = r_A &= \frac{E}{D + E} r_E + \frac{D}{D + E} r_D \\ &= \left(\text{Fraction of firm value financed by equity} \right) \left(\text{Equity cost of capital} \right) + \left(\text{Fraction of firm value financed by debt} \right) \left(\text{Debt cost of capital} \right) \end{aligned}$$

- With perfect capital markets, a firm's WACC is independent of its capital structure and is equal to its equity cost of capital if it is unlevered which matches the cost of capital of its assets.

3. Modigliani-Miller II: Leverage and cost of capital

▪ Levered firm: Capital budgeting and the weighted average cost of capital (WACC)

(a) Equity, debt, and weighted average costs of capital for different amounts of leverage. The rate of increase of r_D and r_E , and thus the shape of the curves, depends on the characteristics of the firm's cash flows.



(a)

(b) Calculating the WACC for alternative capital structures.

E	D	r_E	r_D	$\frac{E}{E+D}r_E + \frac{D}{E+D}r_D$	$= r_{wacc}$
1000	0	15.0%	5.0%	$1.0 \times 15.0\% + 0.0 \times 5.0\%$	$= 15\%$
800	200	17.5%	5.0%	$0.8 \times 17.5\% + 0.2 \times 5.0\%$	$= 15\%$
500	500	25.0%	5.0%	$0.5 \times 25.0\% + 0.5 \times 5.0\%$	$= 15\%$
100	900	75.0%	8.3% ⁴	$0.1 \times 75.0\% + 0.9 \times 8.3\%$	$= 15\%$

(b)

- **Levered firm: Capital budgeting and the weighted average cost of capital (WACC)**
 - With no debt, the WACC is equal to the unlevered equity cost of capital.
 - As the firm borrows at the low cost of capital for debt, its equity cost of capital rises. The net effect is that the firm's WACC is unchanged.

- **Weighted average cost of capital (WACC)**

Reducing Leverage and the Cost of Capital

Problem

NRG Energy, Inc. (NRG) is an energy company with a market debt-equity ratio of 2. Suppose its current debt cost of capital is 6%, and its equity cost of capital is 12%. Suppose also that if NRG issues equity and uses the proceeds to repay its debt and reduce its debt-equity ratio to 1, it will lower its debt cost of capital to 5.5%. With perfect capital markets, what effect will this transaction have on NRG's equity cost of capital and WACC? What would happen if NRG issues even more equity and pays off its debt completely? How would these alternative capital structures affect NRG's enterprise value?

- **Weighted average cost of capital (WACC)**

Solution

We can calculate NRG's initial WACC and unlevered cost of capital using Eqs. 14.6 and 14.7:

$$r_{wacc} = r_U = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D = \frac{1}{1 + 2} (12\%) + \frac{2}{1 + 2} (6\%) = 8\%$$

Given NRG's unlevered cost of capital of 8%, we can use Eq. 14.5 to calculate NRG's equity cost of capital after the reduction in leverage:

$$r_E = r_U + \frac{D}{E} (r_U - r_D) = 8\% + \frac{1}{1} (8\% - 5.5\%) = 10.5\%$$

The reduction in leverage will cause NRG's equity cost of capital to fall to 10.5%. Note, though, that with perfect capital markets, NRG's WACC remains unchanged at $8\% = \frac{1}{2} (10.5\%) + \frac{1}{2} (5.5\%)$, and there is no net gain from this transaction.

If NRG pays off its debt completely, it will be unlevered. Thus, its equity cost of capital will equal its WACC and unlevered cost of capital of 8%.

In either scenario, NRG's WACC and free cash flows remain unchanged. Thus, with perfect capital markets, its enterprise value will not be affected by these different capital structure choices.

- **Computing the WACC with multiple securities**

- If the firm's capital structure is made up of multiple securities, then the WACC is calculated by computing the weighted average cost of capital of all of the firm's securities.

WACC with Multiple Securities

Problem

Compute the WACC for the entrepreneur's project with the capital structure described in Example 14.3.

■ Computing the WACC with multiple securities

Solution

Because the firm has three securities in its capital structure (debt, equity, and the warrant), its weighted average cost of capital is the average return it must pay these three groups of investors:

$$r_{wacc} = r_U = \frac{E}{E + D + W} r_E + \frac{D}{E + D + W} r_D + \frac{W}{E + D + W} r_W$$

From Example 14.3, we know $E = 440$, $D = 500$, and $W = 60$. What are the expected returns for each security? Given the cash flows of the firm, the debt is risk free and has an expected return of $r_D = 5\%$. The warrant has an expected payoff of $\frac{1}{2}(\$210) + \frac{1}{2}(\$0) = \$105$, so its expected return is $r_w = \$105/\$60 - 1 = 75\%$. Equity has a payoff of $(\$1400 - \$525 - \$210) = \665 when cash flows are high and $(\$900 - \$525) = \$375$ when cash flows are low; thus, its expected payoff is $\frac{1}{2}(\$665) + \frac{1}{2}(\$375) = \$520$. The expected return for equity is then $r_E = \$520/\$440 - 1 = 18.18\%$. We can now compute the WACC:

$$r_{wacc} = \frac{\$440}{\$1000} (18.18\%) + \frac{\$500}{\$1000} (5\%) + \frac{\$60}{\$1000} (75\%) = 15\%$$

Once again, the firm's WACC and unlevered cost of capital is 15%, the same as if it were all-equity financed.

■ Unlevered and levered betas

- The effect of leverage on the risk of a firm's securities can also be expressed in terms of beta:

$$\beta_u = \beta_E \frac{E}{E + D} + \beta_D \frac{D}{E + D}$$

- β_u : A measure of the risk of a firm as if it did not have leverage which is equivalent to the beta of the firm's assets.
- Leverage amplifies the market risk of a firm's assets, β_u , raising the market risk of its equity β_E .

$$\beta_E = \beta_u + \frac{D}{E} (\beta_u - \beta_D)$$

- **Betas and leverage**

Betas and Leverage

Problem

Suppose drug retailer CVS has an equity beta of 0.80 and a debt-equity ratio of 0.10. Estimate its asset beta assuming its debt beta is zero. Suppose CVS were to increase its leverage so that its debt-equity ratio was 0.50. Assuming its debt beta were still zero, what would you expect its equity beta to be after the increase in leverage?

- **Betas and leverage**

Solution

We can estimate the unlevered or asset beta for CVS using Eq. 14.8:

$$\beta_U = \frac{E}{E+D}\beta_E + \frac{D}{E+D}\beta_D = \frac{1}{1+D/E}\beta_E = \frac{1}{1+0.1} \times 0.8 = 0.73$$

With the increase in leverage, CVS's equity beta will increase according to Eq. 14.9:

$$\beta_E = \beta_U + \frac{D}{E}(\beta_U - \beta_D) = 0.73 + 0.5(0.73 - 0) = 1.09$$

Thus, CVS's equity beta (and equity cost of capital) will increase with leverage. Note that if CVS's debt beta also increased, the impact of leverage on its equity beta would be somewhat lower—if debt holders share some of the firm's market risk, the equity holders will need to bear less of it.

- **Betas and leverage with cash**

Cash and the Cost of Capital

Problem

In March 2022, Cisco Systems had a market capitalization of \$230 billion. It had debt of \$12 billion as well as cash and short-term investment of \$25 billion. Its equity beta was 0.95 and its debt was approximately zero. What was Cisco's enterprise value at time? Given a risk-free rate of 2.75% and a market risk premium of 5%, estimate the unlevered cost of capital of Cisco's business.

- **Betas and leverage with cash**

Solution

Because Cisco had \$12 billion in debt and \$25 billion in cash. Cisco's net debt = \$12 - \$25 billion = -\$13 billion.

Its enterprise value was therefore \$230 billion - \$13 billion = \$217 billion.

Given a zero beta its net debt, Cisco's unlevered beta was

$$\beta_U = \frac{E}{E + D} \beta_E + \frac{D}{E + D} \beta_D = \frac{\$230}{\$217} (0.95) + \frac{-\$13}{\$217} (0) = 1.01$$

and we can estimate its unlevered cost of capital as

$$r_U = 2.75\% + 1.01 \times 5\% = 7.8\%.$$

Note that because of its cash holdings, Cisco's equity is less risky than its underlying business.

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▪ Leverage and earnings per share

Problem

- LVI is currently an all-equity firm. It expects to generate earnings before interest and taxes (EBIT) of \$10 million over the next year. Currently, LVI has 10 million shares outstanding, and its stock is trading for a price of \$7.50 per share. LVI is considering changing its capital structure by borrowing \$15 million at an interest rate of 8% and using the proceeds to repurchase 2 million shares at \$7.50 per share. What is its EPS before and after the repurchase?

▪ Leverage and earnings per share

Solution (I)

- Suppose LVI has no debt. Since there is no interest and no taxes, LVI's earnings would equal its EBIT and LVI's earnings per share without leverage would be:

$$\text{EPS} = \frac{\text{Earnings}}{\text{Number of shares}} = \frac{\$ 10 \text{ million}}{10 \text{ million shares}} = \$1 \text{ per share}$$

- If LVI recapitalizes, the new debt will obligate LVI to make interest payments each year of \$1.2 million.
 - \$15 million x 8% = \$1.2 million

▪ Leverage and earnings per share

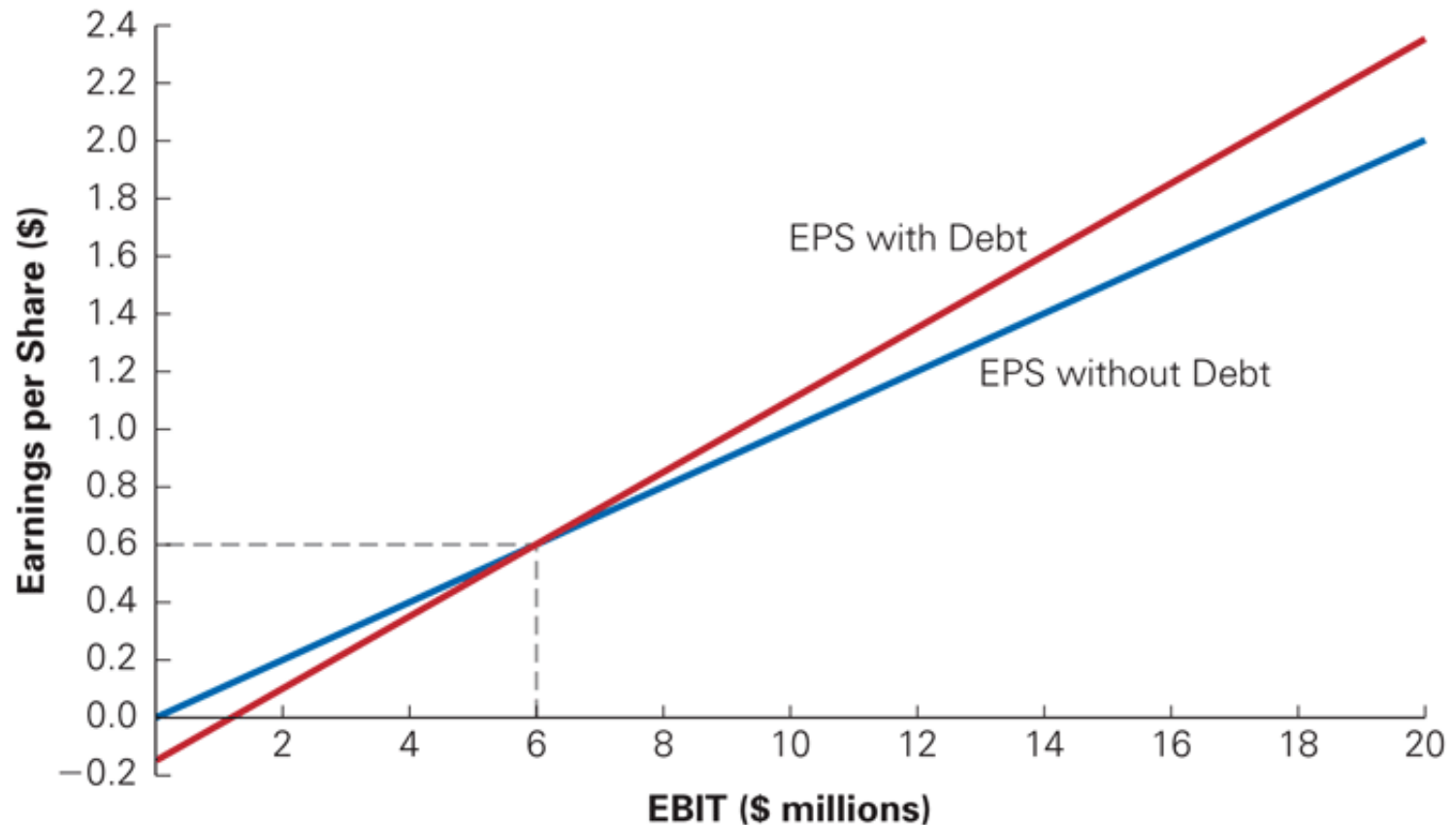
Solution (II)

- As a result, LVI will have expected earnings after interest of \$8.8 million.
 - Earnings = EBIT – Interest
 - Earnings = \$10 million – \$1.2 million = \$8.8 million
- Earnings per share rises to \$1.10.
 - $\$8.8 \text{ million} \div 8 \text{ million shares} = \1.10 per share
 - LVI's expected earnings per share increases with leverage
- Are shareholders better off?
 - No! Although LVI's expected EPS rises with leverage, the risk of its EPS also increases. While EPS increases on average, this increase is necessary to compensate shareholders for the additional risk they are taking, so LVI's share price does not increase as a result of the transaction.

4. Capital structure fallacies

- Leverage and earnings per share

Solution (III)



- **MM propositions and earnings per share**

The MM Propositions and Earnings per Share

Problem

Assume that LVI's EBIT is not expected to grow in the future and that all earnings are paid as dividends. Use MM Propositions I and II to show that the increase in expected EPS for LVI will not lead to an increase in the share price.

■ MM propositions and earnings per share

Solution

Without leverage, expected earnings per share and therefore dividends are \$1 each year, and the share price is \$7.50. Let r_U be LVI's cost of capital without leverage. Then we can value LVI as a perpetuity:

$$P = 7.50 = \frac{Div}{r_U} = \frac{EPS}{r_U} = \frac{1.00}{r_U}$$

Therefore, LVI's current share price implies $r_U = 1/7.50 = 13.33\%$.

The market value of LVI stock without leverage is \$7.50 per share $\times$ 10 million shares = \$75 million. If LVI uses debt to repurchase \$15 million worth of the firm's equity (that is, 2 million shares), then the remaining equity will be worth \$75 million $-$ \$15 million = \$60 million according to MM Proposition I. After the transaction, LVI's debt-equity ratio is \$15 million $\div$ \$60 million = $\frac{1}{4}$. Using MM Proposition II, LVI's equity cost of capital with leverage will be

$$r_E = r_U + \frac{D}{E}(r_U - r_D) = 13.33\% + \frac{1}{4}(13.33\% - 8\%) = 14.66\%$$

Given that expected EPS is now \$1.10 per share, the new value of the shares equals

$$P = \frac{\$1.10}{r_E} = \frac{\$1.10}{14.66\%} = \$7.50 \text{ per share}$$

Thus, even though EPS is higher, due to the additional risk, shareholders will demand a higher return. These effects cancel out, so the price per share is unchanged.

▪ Equity issuances and dilution

- **Dilution:** An increase in the total of shares that will divide a fixed amount of earnings
- **Dilution vs. equity issuance:** It is sometimes (incorrectly) argued that issuing equity will dilute existing shareholders' ownership, so debt financing should be used instead.

▪ Equity issuances and dilution

Problem

- Suppose Jet Sky Airlines (JSA) has no debt and 500 million shares of stock outstanding, currently trading at a price of \$16.
- Last month, the firm announced that it would expand and the expansion will require the purchase of \$1 billion of new planes which will be financed by issuing new equity (62.5 million new shares).

▪ Equity issuances and dilution

Solution (I)

- The current (prior to the issue) value of the equity and the assets of the firm is \$8 billion.
 - 500 million shares x \$16 per share = \$8 billion
- Suppose JSA sells 62.5 million new shares at the current price of \$16 per share to raise the additional \$1 billion needed to purchase the planes.

▪ Equity issuances and dilution

Solution (II)

Assets (\$ million)	Before Equity Issue	After Equity Issue
Cash		1000
Existing assets	8000	8000
	8000	9000
Shares outstanding (million)	500	562.5
Value per share	\$16.00	\$16.00

- The market value of JSA's assets grows because of the additional \$1 billion in cash the firm has raised.
- The number of shares increases:
 - Although the number of shares has grown to 562.5 million, the value per share is unchanged at \$16 per share.

▪ Equity issuances and dilution

- As long as the firm sells the new shares of equity at a fair price, there will be no gain or loss to shareholders associated with the equity issue itself.
- Any gain or loss associated with the transaction will result from the net present value (NPV) of the investments the firm makes with the funds raised.

- **Conservation of value principle for financial markets**

- *With perfect capital markets, financial transactions neither add nor destroy value, but instead represent a repackaging of risk (and therefore return).*
 - This implies that any financial transaction that appears to be a good deal may be exploiting some type of market imperfection.

■ Chapter quiz of sessions 09 + 10

1. How does the risk and cost of capital of levered equity compare to that of unlevered equity? Which is the superior capital structure choice in a perfect capital market?
2. In a perfect capital market, how will a firm's market capitalization change if it borrows capital in order to repurchase shares? How will its share price change?
3. With perfect capital markets, as a firm increases its leverage, how does its debt cost of capital change? Its equity cost of capital? Its weighted average cost of capital?
4. If a change in leverage raises a firm's earnings per share, should this cause its share price to rise in a perfect market?